POLICY FORUM

ARTIFICIAL INTELLIGENCE

GPTs are GPTs: Labor market impact potential of LLMs

Research is needed to estimate how jobs may be affected

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We propose a framework for evaluating the potential impacts of large-language models (LLMs) and associated technologies on work by considering their relevance to the tasks workers perform in their jobs. By applying this framework (with both humans and using an LLM), we estimate that roughly 1.8% of jobs could have over half their tasks affected by LLMs with simple interfaces and general training. When accounting for current and likely future software developments that complement LLM capabilities, this share jumps to just over 46% of jobs. The collective attributes of LLMs such as generative pretrained transformers (GPTs) strongly suggest that they possess key characteristics of other "GPTs," general-purpose technologies (*1, 2*). Our research highlights the need for robust societal evaluations and policy measures to address potential effects of LLMs and complementary technologies on labor markets.

We consider the progress of LLMs' capabilities and the potential breadth of the complementary technologies that they spawn, underscoring that maximizing the impact of LLMs requires their integration within broader systems (*3–5*). The rubric that we develop [see supplementary materials (SM) section 3.1.2] defines task exposure to LLMs, in the spirit of prior work on quantifying exposure to machine learning (*6–8*). Following prior literature, we use the concept of exposure as a proxy for potential economic impact, irrespective of labor-augmentation or displacement effects (see section 2 of the SM for further discussion of the literature). Our approach differs from the broader scope of complementary innovations in general-purpose technology discussions, focusing more narrowly on advanced software capabilities than on the potential for business process reengineering, new intangible assets creation, or workforce retraining. General-purpose technologies such as electricity or computing historically have had far-reaching effects that took decades to fully materialize. With evidence of the general-purpose technology potential of LLMs, we urge caution in making long-term predictions while offering an outline of where work might change.

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RATING TASKS AND OCCUPATIONS

We use the O*NET 27.2 database (*9*), which covers 1016 occupations and their Detailed Work Activities (DWAs) and tasks, focusing on the 923 occupations with available task information.

DWAs describe broader activities that represent the work performed within an occupation, e.g., a DWA of "Execute sales or other financial transactions" might be associated with a range of occupations. Tasks are more specific work units linked to one or more DWAs, such as the task "Deliver email confirmation of completed transactions and shipment." O*NET categorizes tasks as "Core" or "Supplemental," with the latter weighted half as much as the former in our analysis (or noted otherwise). Results are robust to alternative weightings (SM section 9.3). We sourced employment and wage data from the 2020–2022 Occupational Employment series by the US Bureau of Labor Statistics (BLS).

Although the term "LLM" can refer to a broad array of data modalities, we confine our definition of LLM-powered software to principally apply to text, code, and images. We do not tie this analysis to a particular model or model provider but rather to a paradigm of release (through chatbot and application programming interfaces) that was dominant in mid-2023. Progress has been made integrating LLMs with robotics, but the uncertainty around their reliability and adoption, as well as the difficulty conveying the potential of these integrations in our rubric format, led us to exclude assumptions about future robotics integrations from our analysis.

We collect results by applying our exposure rubric (SM section 3.1), where we define exposure as the capacity of an LLM or LLM-powered system to reduce the time required for a human to complete a task by at least 50% while preserving or improving quality. The chosen 50% time reduction threshold, though arbitrary, was used for practical purposes to facilitate interpretation by human annotators and GPT-4. Our estimates of task-level potential impacts are likely to be greater than real-world productivity gains, which will be influenced by factors other than the capabilities of LLMs and complementary technologies (*10*).

Our analysis uses four key measures to gauge exposure levels at the occupation level: (i) E1, the share of an occupation's tasks where access to an LLM alone or with a simple interface would lead to 50% time savings; (ii) E2, the share of an occupation's tasks where additional software is needed on top of an LLM to realize 50% time savings; (iii) E1 + E2, the share of tasks exposed assuming full LLM and software integration; and (iv) E1 + 0.5 × E2, an intermediate measure of exposure while complementary technologies are not fully implemented.

The authors, along with a group of contractors experienced in evaluating LLM outputs, labeled the exposure of DWAs and tasks in the O*NET dataset according to the rubric. We ask human annotators to use their best judgment in labeling DWAs and tasks as E0, E1, or E2 accord-

Human and GPT-4 ratings

Mean and standard deviation of ratings from human and GPT-4 labels are highly similar. Occupation level represents the average share of tasks within an occupation exposed. Task level represents the share of tasks across all occupations exposed.

Occupation-level exposure

	Human		GPT-4	
	Mean	SD	Mean	SD
E1	0.14	0.14	0.14	0.16
E1 + 0.5 × E2	0.30	0.21	0.34	0.22
E1 + E2	0.46	0.30	0.55	0.34

Task-level exposure

	Human		GPT-4	
	Mean	SD	Mean	SD
E1	0.15	0.36	0.14	0.35
E1 + 0.5 × E2	0.31	0.37	0.35	0.35
E1 + E2	0.47	0.50	0.56	0.50

ing to the rubric. Although these human raters had experience working with contemporary LLMs, they do not have diverse occupational experience. This limits their understanding of O*NET work tasks, occupational workflows, and occupation-specific software tools. Future research aims to broaden the pool of human raters across various occupations to enhance the validation of these exposure scores.

We then prompted an early GPT-4 model to apply a slightly modified version of the rubric given to human labelers. We complement the human ratings with GPT-4 ratings to explore the feasibility of using LLMs to aid in social science research, particularly to scale up processes that might otherwise be time-consuming or resource intensive. The version given to GPT-4 was edited to enhance agreement with human ratings across a sample of O*NET task/occupation pairs. Indeed, small changes to the rubric can substantially affect GPT-4's labels, highlighting the fragility of this method in the absence of an alternative validation set (see SM section 3.3.2 for additional discussion of the limitations of this approach). Final agreement rates across human and GPT-4 ratings are visualized at the occupation level in fig. S4 and at the task level in fig. S7.

ESTIMATING EXPOSURE

Our results show LLMs' relevance to approximately 14% of tasks per occupation on average (see the table). When considering the partial implementation of complementary software (E1 + 0.5 × E2), this figure doubles to 30 to 34%, and with full implementation (E1 + E2), it climbs to 46 to 55%. Using the National Employment Matrix data from the BLS, we estimate that ~80% of workers are in occupations with at least 10% of tasks exposed assuming partial implementation of complementary software (E1 + 0.5 × E2), whereas 18.5% of workers have more than 50% of their tasks exposed (fig. S2).

Exposure is prevalent in occupations that involve generating written text or code, as well as those with routine information processing tasks. In fig. S8, we plot exposure by Job Zone—a measure in the O*NET database that groups occupations that are similar in (i) the level of education needed to get a job in the occupation, (ii) the amount of related experience required to do the work, and (iii) the extent of on-the-job training needed to do the work. The amount of preparation required ranges from 3 months (Job Zone 1) to 4 or more years (Job Zone 5). Job Zones 4 and 5 are the most exposed, indicating that occupations needing "extensive preparation" (e.g., lawyers, pharmacists, database administrators) are more exposed than those with lower entry barriers (e.g., dishwashers, floor sanders).

Tasks marked E2 indicate that they are exposed only by software leveraging LLMs and domain-specific scaffolding, not by generally available LLMs. Within an occupation, on average, the addition of domain-specific software exposes an additional 32 to 41% of tasks according to human and GPT-4 ratings, respectively (see occupation-level means in the table). This suggests that LLM-enhanced software could more than double the share of tasks exposed relative to the baseline of LLMs alone (mean E1 of 14% of tasks for both annotation types).

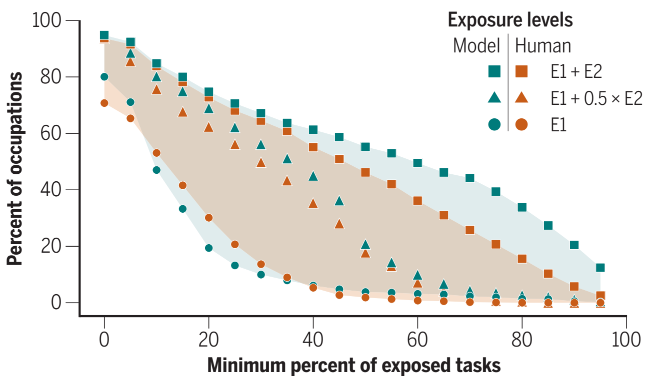
Occupational exposure to LLMs are shown in terms of the percentage of tasks affected (occupational exposure) with specific exposure levels (E1, E1 + 0.5 × E2, E1 + E2) (see the first figure). For instance, human annotators determined that 1.8, 17.8, and 46.2% of occupations are $E1_{50}$, $(E1 + 0.5 \times E2)_{50}$, and $(E1 + E2)_{50}$ exposed (e.g., have at least 50% of tasks exposed at each level), respectively. The vertical gap between E1 and E1 + E2 on this figure indicates the potential additional exposure from artificial intelligence (AI) tools beyond direct LLM interactions alone. In general, higher-wage occupations are more exposed to LLMs than lower-wage occupations (see the second figure).

In exploratory analysis, we used GPT-4 to evaluate the automation potential of tasks using an automation-focused rubric. These GPT-4 ratings were not validated with human ratings. This automation rubric (see SM section 3.2) rates tasks on the basis of the proportion of task components that LLM-based systems could autonomously complete with high quality and reliability. It spans five automation categories, from Full Automation—tasks fully manageable by LLMs without human intervention—to No Automation, where LLMs are unable to reliably perform any part of the task independently with high quality.

Our exploratory analysis estimates only 1.86% of tasks could be fully automated by LLMs plus additional software integrations without human oversight. Still, more than 71% of tasks have at least some component that an LLM plus additional software could plausibly complete with high quality (table S1). High LLM exposure in occupations correlates with higher automation potential, with automation scores explaining 55.6% of the task-level exposure variance (table S2). This suggests that occupations with greater automation risk may also have higher augmentation potential (fig. S3). This challenges the notion that it may be possible to predetermine whether LLMs will

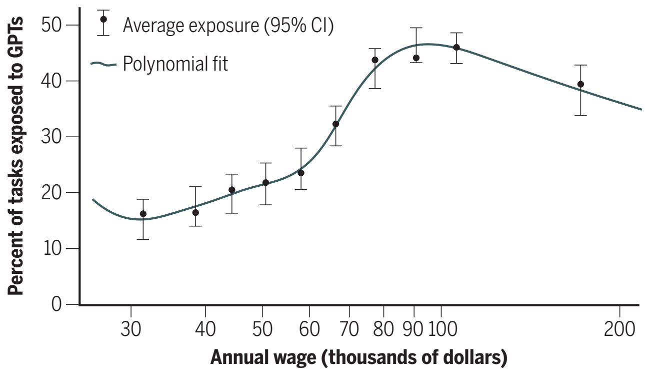
Exposure of occupations to GPTs

Exposure intensity across the economy is shown in terms of percent of affected occupations. The horizontal axis represents the share of tasks within an occupation exposed at the different exposure levels denoted in the figure legend. Shaded areas denote the range of exposure levels.



Exposure to GPTs by income

The share of all tasks within occupations that are exposed to LLMs and partial LLM-powered software (E1 + 0.5 × E2) is shown against the median annual wage for the occupation. Data reflect human rating.



CI, confidence interval; GPT, generative pretrained transformer; LLM, large language model.

GRAPHIC: M. HERSHER/*SCIENCE*

augment or automate jobs, despite recommendations for AI to complement rather than replace human work (*11*).

One specific criterion for general-purpose technologies is that they exhibit innovational complementarities: The productivity of research and development (R&D) efforts in downstream industries improves owing to their development. This could lead not only to capital deepening (an increase in the capital-to-labor ratio), but also a change in the direction of innovation (*3*). Our data can offer some suggestions as to whether work will change in R&D roles as a result of LLMs. We construct a network of occupations using detailed work activities that they share in common (fig. S13). Nodes in this network are O*NET occupational varieties, and nodes share an edge if they share a DWA in common. The network is highly modular (Louvain modularity of 0.709), and we detect 11 distinct occupational clusters using the Louvain method. The two job groups (clusters) that are most exposed to LLMs are "Scientists and Researchers," then "Technologists," such as software engineers and data scientists (fig. S16). The R&D-intensive roles in the top two categories are in general more heavily exposed than many other kinds of work. This suggests that when LLMs improve, they have the potential to cause downstream improvements in R&D productivity for workers in the sectors deploying them (see SM section 8.2.).

IMPLICATIONS AND LIMITATIONS

As LLMs are nascent, public investment in tracking LLM adoption and ensuing labor market impacts is urgently needed and will be vital for shaping effective policy responses. Although this paper does not test specific policies, recent work has outlined US policy options on education, training, tax, and safety net reforms that could help ensure that the economic gains from LLMs are large and broadly distributed (*12*). Transitional policy tools are essential to preventing harms (e.g., from concentrated labor demand shocks) for negatively affected workers. Initiatives such as wage insurance and unemployment insurance reform can ease transitions and bolster economic security. Furthermore, collaborations between AI labs and governments to extend AI's economic gains equitably could help to direct the development of LLMs in a more socially beneficial direction.

Despite a growing literature that aims to understand the labor market impacts of these systems, there is still no clear understanding of how "exposure" to AI systems will translate to real-world impacts on labor demand, wages, inequality, job quality, and other key outcomes. Public investments in measurement and tracking of LLM adoption, complementary technology development, and the ensuing labor market impacts are a critical input to informing optimal public policy responses.

Using a rubric-based approach to measure exposure has limitations (SM section 3.3). It does not capture the creation of potential new tasks and potential changes to returns to technological capital (*13*). Furthermore, we cannot conclude that there will be any impacts related to the equilibrium demand for human labor across occupations using this method. This method also misses shifts in innovation direction rather than just capital deepening (*3*), even if it can suggest where LLMs might influence labor task intermediates to innovation. Despite these shortcomings, this measurement is useful for identifying tasks and occupations likely to be affected by LLMs. However, predicting long-term labor market changes due to LLMs' general-purpose abilities remains a complex challenge.

There are specific limitations in using a LLM to assist with this rubric-based methodology. Task classification by GPT-4 is affected by changes in rubric language, prompt structure, example inclusion, detail level, key term definitions, and model version. Refining prompts with a small validation set can better align model outputs with the rubric's intended meaning. We made slight adjustments between GPT-4 and human annotator rubrics to guide the model toward accurate labels without biasing human judgments. Although we use multiple annotation sources, none is considered the definitive ground truth. There is room for improvement and innovation in creating effective rubrics for LLM classification of complex information.

We do not systematically examine sources of disagreement across labeling sources, but there are a few categories of tasks that qualitatively appear to have greater variation in exposure ratings. These include tasks where using an LLM or domain-specific software would necessitate substantial individual or group behavior or norm changes (such as expecting to meet with humans to conduct a meeting or negotiation); tasks where human oversight, judgment, or empathy are expected or required (such as certain kinds of decision-making or counseling); and tasks already potentially exposed by technology where it is difficult to judge the added impact of using an LLM (such as making reservations).

Predicting the trajectory of LLM applications is fraught with uncertainties owing to emergent capabilities, shifts in human biases, and technological evolution. Our forward-looking projections, rooted in contemporary trends and perceived technological potential, are subject to change with new breakthroughs. Tasks currently deemed beyond the reach of LLMs may become viable with future innovations, whereas those seeming ripe for disruption could encounter unexpected barriers. Through continued effort to better understand LLM capabilities and their possible effects on the workforce, policymakers and other key stakeholders can make better-informed decisions, efficiently navigating the complex landscape of AI and its implications for the future of work. ■

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SUPPLEMENTARY MATERIALS

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